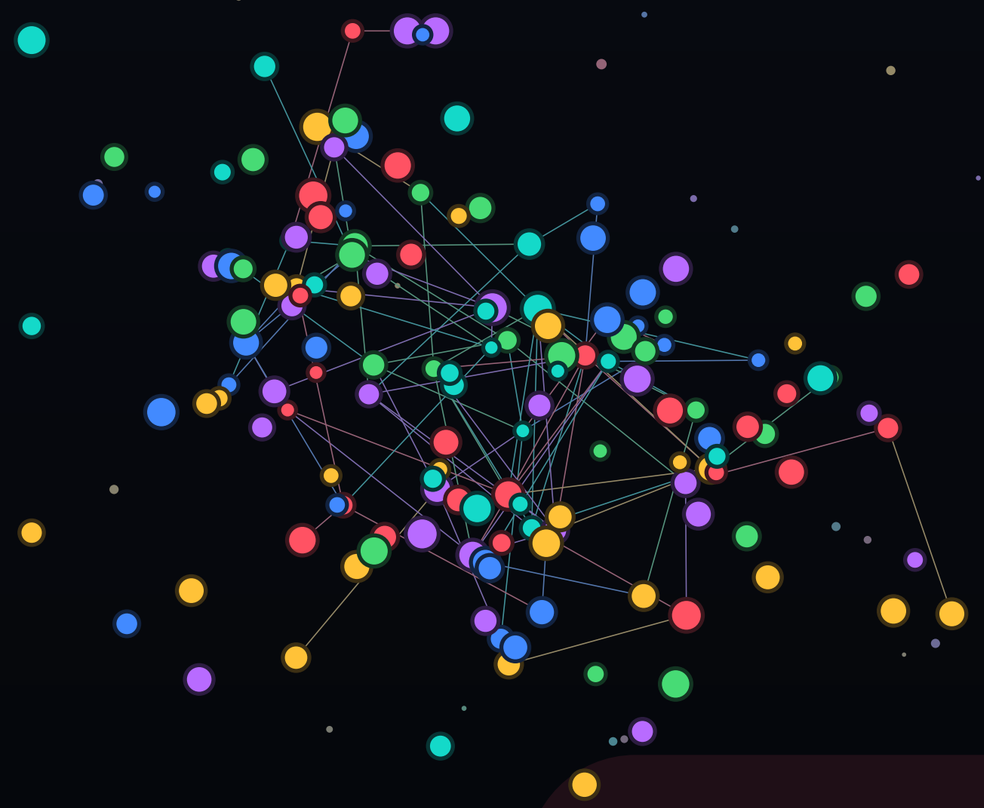
FROM RULES TO WORLDS

How simple pushes become complex motion

Particle life is a useful way to show emergence: local interactions creating global structure.



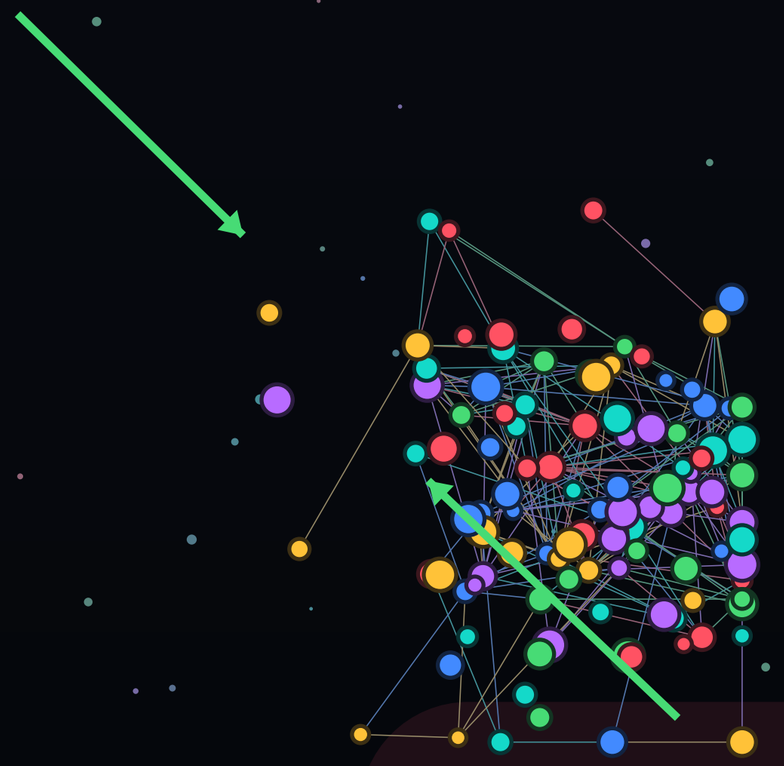
thousands of local interactions

ATTRACTION

Pull creates groups

When two species attract, they gather. Enough attraction creates islands, strands, and orbiting clumps.

- Weak pull makes soft clouds.
- Strong pull makes tight centers.
- Mixed pull makes layered groups.

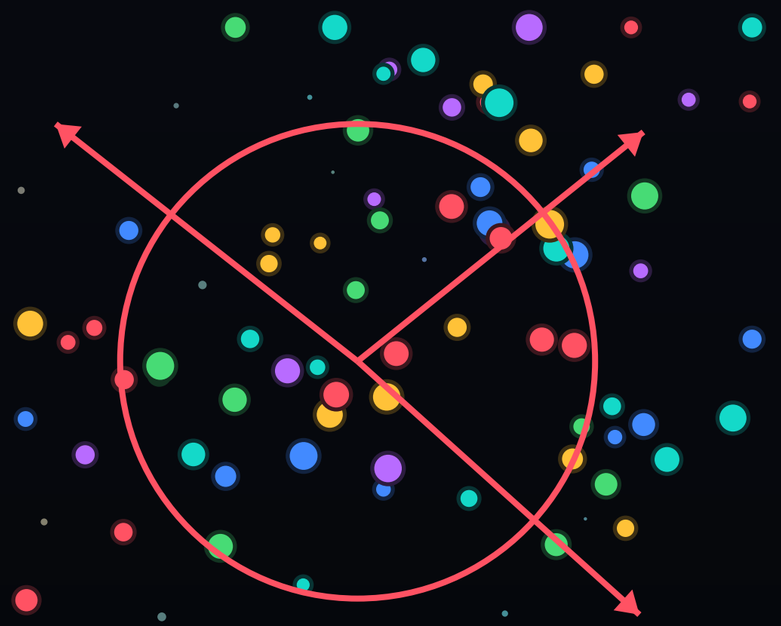


REPULSION

Push creates space

Repulsion is just as important. It prevents collapse and creates boundaries between moving groups.

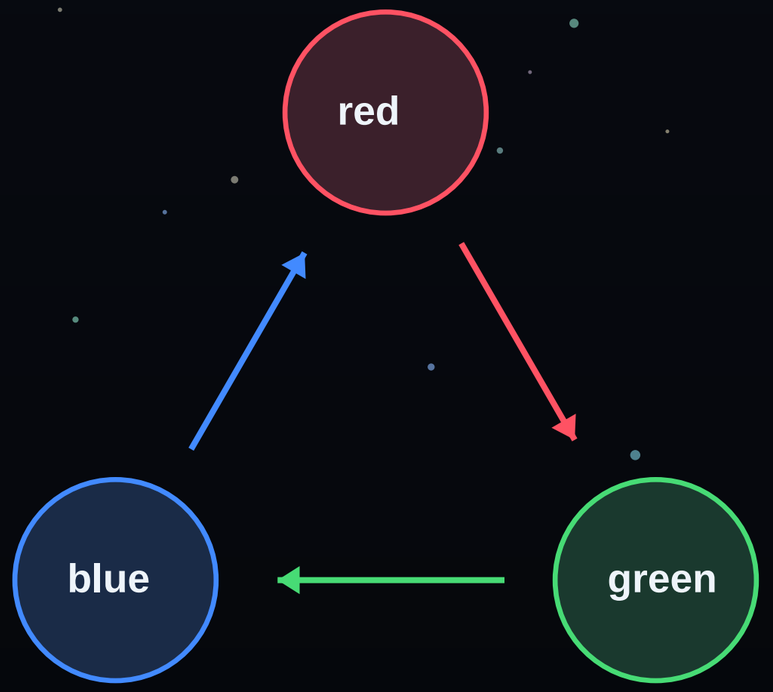
- Short-range push keeps particles separated.
- Longer repulsion can carve lanes.
- Balance matters more than any one force.



FEEDBACK

Cycles create chase behavior

If red follows green, green follows blue, and blue avoids red, the system can start to chase itself.



SPACE

The world wraps around

Particles that leave one side re-enter from the other. That keeps the simulation dense and continuous.

Wrapped world

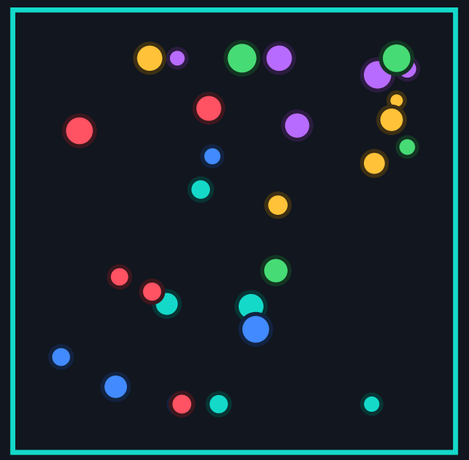
edges connect, so motion never falls off the map

DIMENSIONS

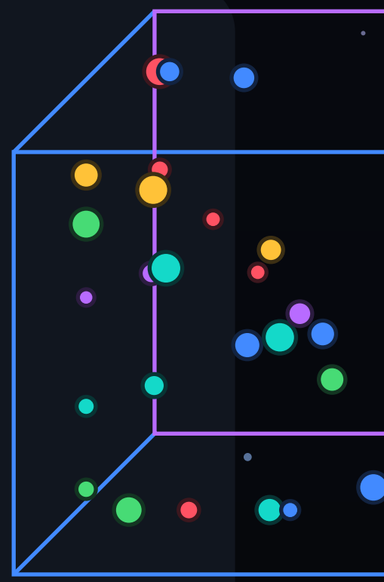
2D is readable. 3D is spatial.

The rule is the same in both modes. The difference is how much room the particles have to arrange themselves.

2D and 3D are the same idea



2D plane



3D volun

WATCH FOR

The story is in the changing shapes

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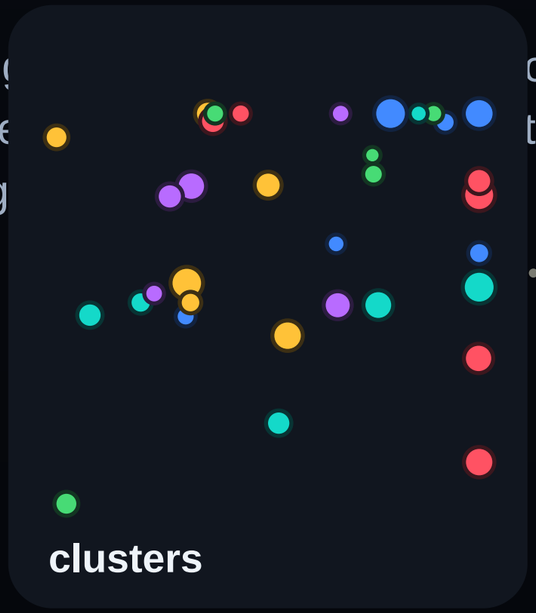
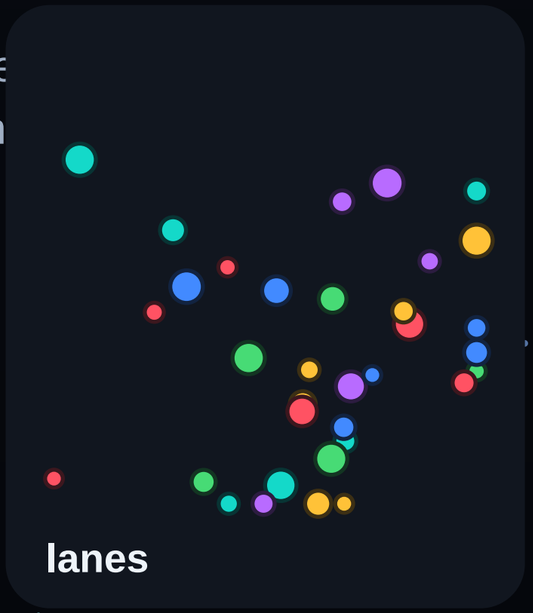
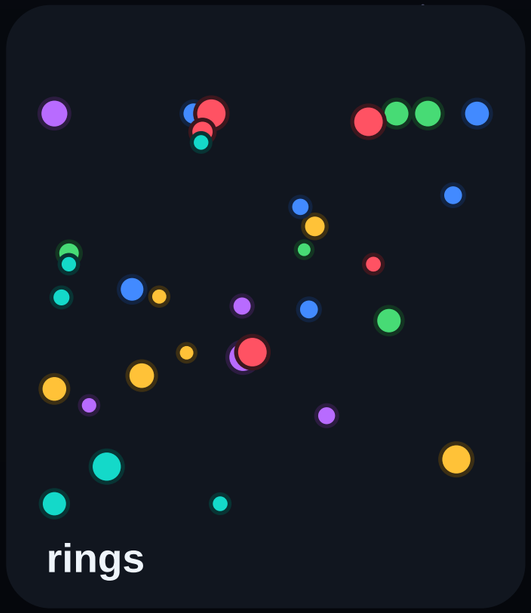


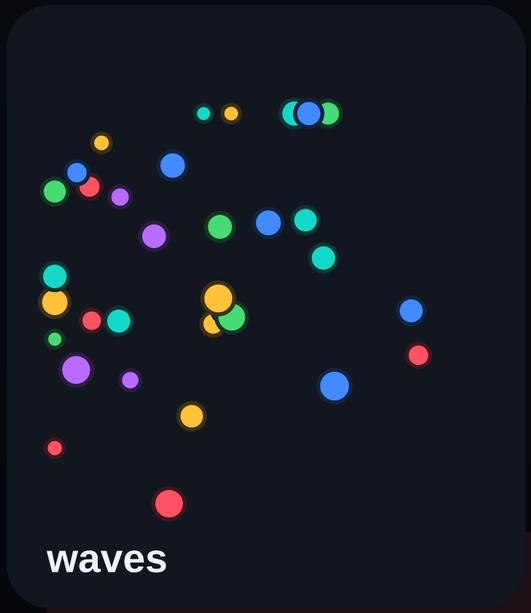
clusters



lanes



rings



waves